LESTO – LEad fast reactor Safety design and TOols

## Deliverable D38

# The coupled GEMS and CFD codes for fission products reactive transport modelling

Description of the developed coupling and modelling

Jan Kren
Paul Scherrer Institute (PSI), 5232 Villigen PSI, Switzerland

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## Executive summary

This deliverable describes the coupling of the OpenFOAM finite-volume solver `rhoFixedFlowFoam` with the GEMS3K Gibbs-energy minimisation kernel [1], developed at PSI for the reactive transport of volatile fission-product species. The test bed is the thermal-gradient (thermosublimatography) column, in which iodine compounds evaporated from lead–bismuth eutectic (LBE) are carried by helium along a falling wall temperature and deposit by volatility [2, 3].

**What was developed.** (i) A transport solver for any number of trace gases in a frozen helium carrier, with reversible Hertz–Knudsen–Schrage (HKS) wall exchange in series with the near-wall diffusion resistance, shared condensed wall inventories, a reactive BiI wall channel and an exact element ledger for Pb, Bi and I. (ii) A plain-C bridge to GEMS3K that links a static GEMS3K into the OpenFOAM executable. (iii) Two coupling points: GEMS3K equilibrium pressures for the HKS law, and homogeneous gas re-speciation of nine Pb–Bi–I gases per cell by three engines: an exact ideal-gas element-potential kernel, GEMS3K standard states with that kernel (*frozen*), and warm per-cell GEMS3K calls refined by the kernel (*local*). (iv) GEMS3K input systems for Pb–I–He and Pb–Bi–I–He generated from thermodynamic records without GEM-Selektor.

**What was established.** The coupled code conserves elements to round-off: the largest raw element residue of the 19 production runs is $1.41 \times 10^{-12}$ of the supplied material. On a qualification channel the frozen engine reproduces the kernel deposits within $1.1 \times 10^{-7}$ (L1) and the local engine reproduces the frozen one exactly; 5,814,420 local calls cost 132 $\mu$s each on average. Mapping GEMS3K over 321,480 Pb–Bi–I and Pb–I calls showed that its interior-point method stalls in hot, non-dilute Bi-iodide gas and is silently inaccurate for trace species; the coupling therefore guards every call and refines or replaces it by the exact kernel. The production results reported here use the exact kernel with formation constants generated from the same standard-state data as the GEMS3K system; a full-size per-cell GEMS3K run of the full production case (41,472 cells, 60 s) reproduces the kernel deposits within $2.1 \times 10^{-4}$ (L1), the order of parallel round-off; its 514.5 million GEMS3K calls all succeeded, and the final speciation is computed by the kernel from GEMS3K's thermodynamic data (section 4.5).

**Modelling results.** Applied to Liu et al. (2025) [2], the model places all five computed binned iodide peak temperatures inside the printed experimental ranges; the comparison is conditional, because the source masses are taken from the same deposits. The computed $BiI_3/PbI_2$ split depends strongly on BiI(g) thermodynamics: a $+29\,\mathrm{kJ\,mol^{-1}}$ surrogate shift changes the $BiI_3$ share of deposited iodide iodine from 94.58% to 2.01%. Without tuning, the same wall law reproduces the deposition-onset temperatures of five $PbI_2$ experiments of Lobresco et al. (2026) [3] with an RMS deviation of 18 K, against 35 K for both models of that study. These are verification and conditional comparisons, not a validation of the model.

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# 1 Introduction

**Purpose within Task 3.4.** Task 3.4 of LESTO addresses the release and transport of fission products from the coolant source. The approach of PSI combines computational fluid dynamics (CFD) with chemical-thermodynamic equilibrium computed by GEMS/GEMS3K [1, 4], following earlier CFD–GEM coupling for reactor coolant chemistry [5]. This deliverable documents the coupled code as built: its equations and data flow, the coupling interface, its verification, and its first modelling applications.

**The physical problem.** Iodine released from LBE can travel as several gaseous species (Pb, PbI, $PbI_2$, Bi, $Bi_2$, BiI, $BiI_3$, I, $I_2$) before it deposits on cooler surfaces. The chemical form sets both the gas inventory available for transport and the equilibrium pressure that governs wall exchange. Thermal-gradient columns separate volatile species along a falling wall temperature and provide spatially resolved deposition data [2, 3, 6, 7]. In fluid-dynamic terms, a column is a slow laminar helium flow (Re $\approx$ 0.7–3) in a narrow tube, where axial advection, radial diffusion to the wall and a nonlinear, temperature-dependent wall exchange together set the deposit profile (fig. 2).

**What D38 delivers.**

- The CFD solver `rhoFixedFlowFoam` for OpenFOAM v2412 and v2606, with multi-species transport, wall exchange and element accounting (section 2).
- The GEMS3K coupling: bridge library, input-system generator, coupling points, guards and fallbacks, with measured robustness and cost (section 3).
- A regression and verification suite and the numerical studies of the production cases (section 4).
- Two modelling applications, Liu et al. (2025) and Lobresco et al. (2026) (section 5), and a reproducible case ladder (section 6).

A methods manuscript based on the same results is in preparation; this report reuses its figures and generated tables.

# 2 The CFD code: rhoFixedFlowFoam

**Frozen carrier.** The carrier fields $\rho$, $\mathbf{u}$, $T$, $p$ and the face mass flux $\phi$ are taken from a converged steady laminar helium calculation with `rhoSimpleFoam` and are held fixed. A flux-potential projection removes residual discrete divergence before species transport, and continuity is checked to $10^{-12}$ of the inlet flow. No momentum, pressure or energy equation is solved during species transport, so the trace species do not feed back on the carrier.

**Species transport.** Each gas $i$ is carried as a mass ratio $Y_i$ and satisfies

$$\frac{\partial(\rho Y_i)}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = \nabla \cdot (\rho D_i \nabla Y_i) + S_i^{\mathrm{rel}} + S_i^{\mathrm{wall}}, \tag{1}$$

with implicit Euler time integration, a run-time selectable advection scheme and constant, $PbI_2$–He or Chapman–Enskog diffusivities. Species are defined at run time; adding a species needs no recompilation. Each species carries a chemical formula, checked against its molar mass, from which the element concentrations $b_k = \sum_i \nu_{ki} \rho Y_i / M_i$ ($k \in \{\mathrm{Pb}, \mathrm{Bi}, \mathrm{I}\}$) are formed.

**Wall exchange.** A gas is paired with a condensed species stored as an immobile inventory on each wall face (an *element*). The exchange law is affine in the cell value, $q_e = a_e - g_e Y_c$, and is assembled implicitly. Two laws are implemented:

- the *temperature model*, a temperature-activated deposition velocity $v_d[1 - e^{-k(T_{\mathrm{dep}} - T_e)}]^+$, as in the temperature-based model of Lobresco et al. [3];
- the *HKS law* [8–10], $j_i = G_i[p_{\mathrm{eq},i}(T_w) - p_{i,w}]$ with $G_i = C_e s_i \frac{2\alpha_i}{2-\alpha_i}\sqrt{M_i/(2\pi R T_w)}$, placed in series with the diffusion over the cell-centre-to-wall distance $d_f$:

$$G_{i,f}^{\mathrm{eff}} = \frac{G_i}{1 + G_i \chi_{i,c} d_f/(\rho D_i)_f}, \qquad j_{i,f} = G_{i,f}^{\mathrm{eff}}\left[p_{\mathrm{eq},i}(T_w) - \chi_{i,c} Y_{i,c}\right], \tag{2}$$

  with $\chi_{i,c} = \rho_c R T_c / M_i$.

The equilibrium pressure $p_{\mathrm{eq}}$ comes from phase-resolved vapour-pressure tables or from GEMS3K (section 3.2). Evaporation is limited by the face inventory over the step, and inventory updates use the exchange realised by the final gas solve, so switching between deposition and evaporation cannot create material. Further channels connect Pb(g) to condensed Pb, Bi(g) and $Bi_2$(g) to one shared condensed Bi inventory, and the irreversible reactive channel $3\,\mathrm{BiI}(g) \rightarrow 2\,\mathrm{Bi}(c) + \mathrm{BiI_3}(g)$ with $p_{\mathrm{eq,BiI}} = p^\circ[(p_{\mathrm{BiI_3}}/p^\circ)/K_r(T_w)]^{1/3}$.

**Sources and inventories.** Releases are prescribed per species (source model Q1), as elements plus a metal-vapour fraction of saturation $\beta$ (Q2), or as evaporating inventory samples (Q3); samples can be removed at a given time.

**Element ledger.** Every store (gas, wall, sample, outflow, removed, released) is booked per pair, per species and per element in binary restart records. For each element the solver reports the raw residue

$$R_k = N_k^{\mathrm{gas}} + N_k^{\mathrm{wall}} + N_k^{\mathrm{sample}} + N_k^{\mathrm{net\,out}} + N_k^{\mathrm{removed}} - N_k^{\mathrm{initial}} - N_k^{\mathrm{released}}, \tag{3}$$

and books the linear-solver defect and any clamping separately, so that the bookkeeping closure and the raw material balance can both be checked at every step.

# 3 The GEMS3K coupling

## 3.1 Architecture

**Why a static C bridge.** GEMS3K 4.6.1 requires C++20, which the system compiler of the OpenFOAM v2412 installation (gcc 7.5) cannot compile. Linking the shared GEMS3K library directly into an OpenFOAM executable fails because two libstdc++ versions meet in one process (an unresolved symbol of version `CXXABI_1.3.15`). The bridge `thermochemistry/gemsbridge` (`libgemsbridge.so`) avoids this: GEMS3K, libstdc++ and libgcc are linked statically with hidden symbols, a version script exports only the `gemsb_*` C functions, and the library depends on libc and libm only. It exports 39 functions: the 30 original ones and nine additions of the gas-engine stage (standard-state $G^\circ/RT$, element potentials, gas mole fractions, dual pair equilibrium pressure, condensate suppression, element-balance error, IPM control and a warm iteration cap). `make check` rejects any further export or runtime dependency, and 800 cold and warm calls of the original interface are bit-identical before and after the extension. There is one GEMS3K engine per MPI rank and no MPI inside the bridge.

**Build options.** The default solver build is bridge-free. Setting `LESTO_GEMSBRIDGE` to the bridge directory compiles the coupling in (generated header `gemsConfig.H`, with automatic clean rebuilds when the state changes); a bridge-free build stops with a rebuild hint if a GEMS engine is selected. A separate option, `LESTO_XGEMS`, compiles an xGEMS start-up call inherited from the main development branch; it only equilibrates a demonstration system and is not part of the coupling described here.

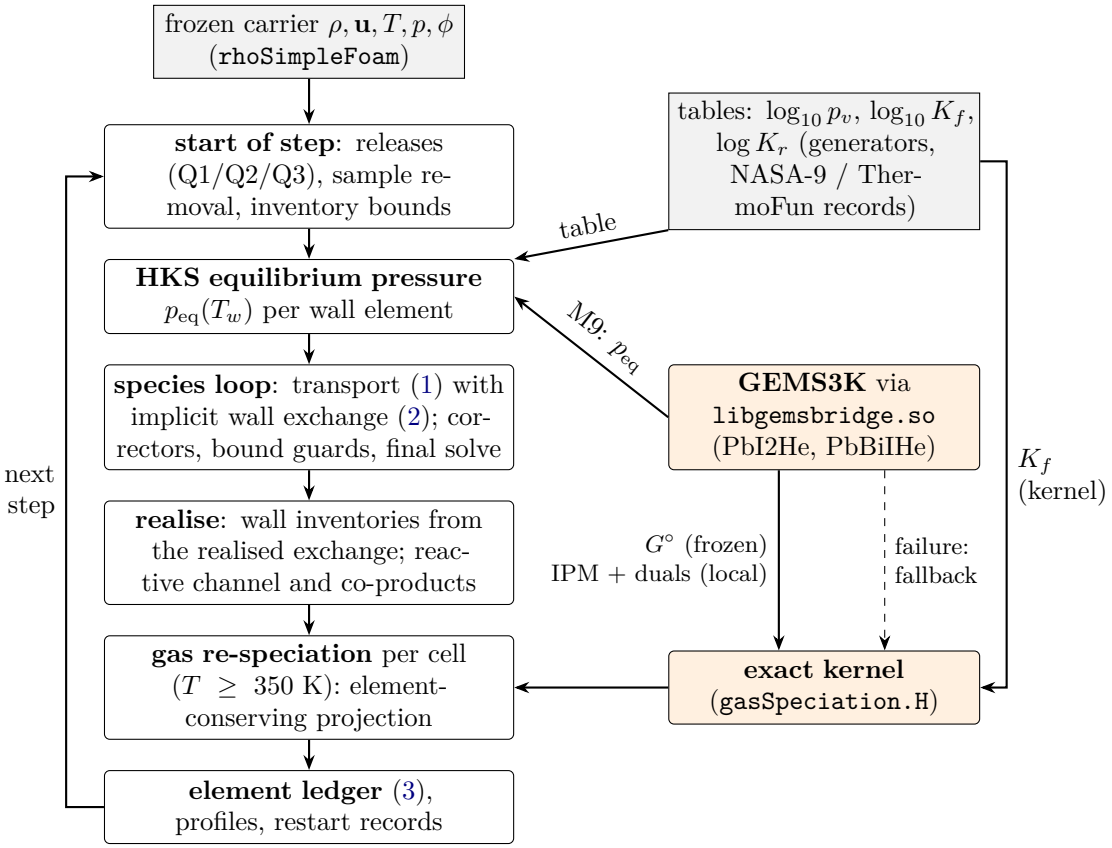



**Figure 1:** Data flow of one time step of `rhoFixedFlowFoam` and its two GEMS3K coupling points. Left: operator sequence of the solver. Right: thermodynamic inputs. The HKS equilibrium pressure comes from a table or, optionally, from GEMS3K (M9). Gas re-speciation always ends in the exact kernel; GEMS3K supplies either its standard states (mode frozen) or a warm interior-point (IPM) result whose element potentials start the kernel refinement (mode local). Dashed: fallback after a failed or guarded GEMS3K call.

**GEMS3K systems.** A generator (`make_gems3k_input.py`) writes the GEMS3K input set (DCH, IPM, DBR and `.lst` files) without GEM-Selektor: one ideal-gas phase plus pure condensed phases, with $G^\circ(T)$ from NASA-9 records [11] or ThermoFun records [12].

- **PbI2He**: He, I, $I_2$, Pb, PbI, $PbI_2$ (g); $PbI_2$(cr,l), Pb(cr,l); Gurvich data as distributed with NASA CEA (Apache-2.0).
- **PbBiIHe**: 4 elements, 19 species, 10 phases; grid 270–1250 K every 10 K and 0.8–1.2 bar. The ten species shared with PbI2He are bit-identical. Bi–I data derive from HERACLES (Barin [13]) and are external inputs (section 6).

Checked through the bridge, GEMS3K's gas speciation gives the equilibrium constants of the Bi–I reactions of its own records to $\leq 2.5 \times 10^{-4}$ decades, and run-time ThermoFun evaluation agrees with the tabulated grid to $\leq 1.2 \times 10^{-4}$ decades. The bridge also guards three GEMS3K pitfalls found in testing: the temperature tolerance below which $G^\circ$ is not re-interpolated (clamped to $10^{-3}$ K), hard errors for element amounts below $10^{-17}$ mol (normalisation and a $10^{-15}$ floor), and stalled calls that report success (stall acceptance disabled, and both status and mass balance checked).

### 3.2 Coupling point 1: equilibrium pressure of the HKS law

For a gas–condensate pair, the solver calls GEMS3K with every condensed species suppressed; GEMS3K then returns the homogeneous gas at the bulk composition, and the dual activity of a suppressed condensate is its saturation ratio $\Omega$. The equilibrium pressure is $p_{eq} = p_g 10^{-\max_\varphi \log_{10} \Omega_\varphi}$ over the pair's condensates. Two modes exist:

- **frozen**: cold calls at the table nodes at start-up, which replace the table values (PbI2He: 152

nodes in 500–1250 K, 0 failed, 0 rejected, largest deviation from the tabulated vapour pressure $5.7 \times 10^{-5}$ decades); deterministic and restart-exact;

- **local**: one call per wall element at the start of each step, at the element's temperature and its cell's composition, warm started from the element's own previous state. Guards fall back to the table below 500 K, below a mole fraction of $10^{-6}$, outside the GEMS3K grid, after a failed call and for results more than 0.01 decades from the table; a fatal error re-creates the engine. The largest deviation from the table in all M9 runs was $3.8 \times 10^{-4}$ decades; on the $PbI_2$ system a warm call cost about 60 $\mu$s (3.9 IPM iterations) with no failures in a CFD-like test loop over 1000 wall cells.

For Bi systems the pressure is taken from the element potentials (dual form), which agrees with the vapour-pressure tables to $8.37 \times 10^{-5}$ decades across 64 values, with additional guards (section 3.5). For an ideal gas over a pure condensate of the same formula, $p_{\mathrm{eq}}$ depends on $T$ only, so the table is exact and GEMS3K adds no information; the Pb–Bi–I production cases therefore use tables generated from the same thermodynamic records. GEMS3K pressures become necessary for non-ideal or multi-component condensed phases, which are not yet modelled.

### 3.3 Coupling point 2: homogeneous gas re-speciation

After the species loop, every cell above 350 K is projected onto the homogeneous ideal-gas equilibrium of its element amounts: find the partial pressures $p_i$ and element potentials $\lambda_k$ with

$$\frac{p_i}{p^\circ} = K_{f,i}(T) \exp\Big(\sum_k \nu_{ki}\lambda_k\Big), \qquad \sum_i \nu_{ki}\frac{p_i}{RT} = b_k, \qquad p^\circ = 10^5\,\mathrm{Pa}, \tag{4}$$

and return $Y_i = M_i p_i/(\rho RT)$. Condensed phases are suppressed in this step; condensation happens only through wall exchange. The engine is selected in `respeciation{engine...}`:

| Engine | Thermodynamic data | Solver per cell and step |
|---|---|---|
| `kernel` | $\log_{10} K_f(T)$ table generated from the NASA-9 / ThermoFun records | exact element-potential Newton solver (`gasSpeciation.H`); no bridge needed |
| `GEMS`, `frozen` | GEMS3K standard states, cached per unique cell $(T, p)$ at start-up | the same exact kernel |
| `GEMS`, `local` | GEMS3K | guarded, capped warm IPM call; valid element potentials start the kernel refinement; canonical cold final refinement; kernel fallback on failure |

The kernel works in logarithmic variables with a safeguarded Newton iteration, a Schur-complement reduction of the Pb and Bi subproblems, an extended-precision fallback for dilute bulks and bisection where Newton steps fail to contract. Over 160,740 stored gas-map inputs it agrees with an independent reference to $1.8 \times 10^{-10}$; a cold call costs 16.2 $\mu$s on the stored 432-point set and 26–30 $\mu$s on the broader map. The GEMS3K and open-data formation constants differ by at most $3.69 \times 10^{-4}$ decades, so the engines differ in their standard-state data at that level and otherwise solve the same equations.

### 3.4 Operator splitting and conservation

The step is a first-order (Lie) split (fig. 1): transport with implicit wall exchange, realisation of the wall inventories, then re-speciation once per step, never inside the corrector loop. The re-speciated amounts are exponentials of converged potentials, absent elements are removed exactly, cells with a non-positive element amount are skipped and counted, and slightly negative $Y$ values are made positive by element-conserving re-speciation rather than clipping. The residue of every transfer between stores is booked as a CONVERSION entry of the element ledger, so closure stays at round-off without drift. Splitting error is first order in $\Delta t$ and is assessed by time-step studies (section 4.3).

**Table 1:** Modelled CPU-hours of a 60 s run at 1 ms on the 228,960-cell pipe of the cost model, for equilibrium calls (a) at the wall elements, (b) at the walls and in cells above 700 K, (c) in every cell. For comparison, transport costs about 40 CPU-h per paired species and 360 CPU-h for nine species.

| Engine | (a) | (b) | (c) |
|---|---|---|---|
| GEMS3K, $PbI_2$-like gas | 17 | 173 | 363 |
| GEMS3K, $PbI_2$ + $BiI_3$ (with failures) | 32 | 319 | 669 |
| GEMS3K, $BiI_3$-rich gas | 94 | 936 | 1960 |
| Exact solver | 3.2 | 32 | 67 |
| Exact solver, $G^\circ$ cached per cell | 1.4 | 13.7 | 28.6 |

### 3.5 Robustness, guards and cost of GEMS3K

Before GEMS3K was admitted per cell, it was mapped over 321,480 calls per run (eight Pb–Bi–I element mixes and $PbI_2$, trace fractions from $10^{-12}$ to 0.32, 95 temperatures in 304–1244 K, one cold and three warm calls per point, condensed species suppressed or allowed), each compared with an exact equilibrium of GEMS3K's own $G^\circ$. Main findings:

- **Failures in hot, non-dilute Bi-iodide gas.** With condensed species suppressed, at $T \geq 500$ K, $BiI_3$ mixes fail in 0.1%, 6%, 25% and 42% of calls at trace fractions $10^{-4}$, $10^{-3}$, $10^{-2}$ and $10^{-1}$; metal-rich LBE-vapour mixes in 73–100% at $10^{-1}$; $PbI_2$ in at most 1%. The interior-point criterion stalls at $7 \times 10^{-5}$–$10^{-4}$ while $G$ has converged, and the call runs to its 7000-iteration limit (about 45 ms). Warm starts can be worse than cold ones (7158 against 27 iterations).
- **The element balance is never wrong.** None of the 311,976 converged calls is off by more than $10^{-10}$, so the balance check cannot detect the silent errors, which follow the species' own mole fraction $x_g$: $\leq 4 \times 10^{-4}$ decades for $x_g \geq 10^{-6}$, but 0.04 at $10^{-8}$ and $\geq 1$ below $10^{-11}$. The primal $p_{\mathrm{eq}}$ can be off by 0.33 decades at a trace fraction of $10^{-8}$, whereas the dual $p_{\mathrm{eq}}$ is exact to $\leq 10^{-9}$ decades in every converged call.
- **An exact solver with GEMS3K's** $G^\circ$ had 0 failures in 321,480 warm-chain and 54,720 cold calls. This motivated the kernel and the frozen/local designs above.

The implemented guards are: minimum temperature and mole fraction, an upper mole-fraction bound on the trace formula units (at $x \leq 3.2 \times 10^{-4}$ the mapped Bi-iodide gas had 0 failed calls, against 8.5% without the bound), the use of speciation factors only for $x_g \geq 10^{-7}$, a warm cap of 200 iterations followed by a cold retry, and dual pressures. Fatal calls re-create the engine and invalidate the warm states; failed calls and NaN duals take the kernel and are counted.

Measured on one thread (64 cells at 600–1000 K, a new $T$ every call, $x = 10^{-4}$), a warm call costs 90.8 $\mu$s with GEMS3K for PbBiIHe (57.2 $\mu$s for PbI2He), 17.0 $\mu$s with the exact solver and 7.5 $\mu$s with the exact solver given the cell's $G^\circ$. A fit gives about 80 $\mu$s plus 6.4 $\mu$s per IPM iteration for GEMS3K, and 1.3–3.5 ms with Bi iodides at $x \geq 10^{-3}$. On a hyperthread whose sibling runs another job the same calls take 2.1–2.5 times as long, so production runs need whole cores. Table 1 scales these costs to a full run. A tabulated surrogate of the Pb–Bi–I equilibrium was evaluated and not recommended: a 4-D table reaches only about $10^{-2}$ decades, the minor species change by factors of 10–1000 across the stoichiometric ridges, and the saving over the cached exact solver is about 26 CPU-h in case (c).

### 3.6 What is and what is not coupled

- **Coupled:** GEMS3K equilibrium pressures for the HKS law (frozen or local); GEMS3K standard states or warm GEMS3K equilibria for the per-cell gas speciation (frozen or local); GEMS3K-generated systems from the same records as the solver's tables.
- **Engine of record for the reported results:** the exact kernel with formation constants and vapour-pressure tables generated from the same standard-state records. The per-cell GEMS3K

engines are a qualified capability (section 4.4); in mode local the IPM result only starts a kernel solve and the final speciation is the canonical cold kernel solve, so local and frozen are bitwise identical (section 4.5); this departs from the original design, in which a polished IPM result was to be used.

- **Not coupled:** condensed-phase equilibrium inside GEMS3K (condensed phases are suppressed; condensation is wall exchange); source equilibrium with liquid LBE (source model Q4), which needs the unpublished iodine activity in LBE; feedback of species or latent heat on the frozen carrier; non-ideal gas or solution phases; aerosol nucleation; surface chemistry with steel or silica.

# 4 Verification

## 4.1 Test suite

`tests/Alltest` runs pass/fail criteria by milestone (M0 regression gate to M10d) and records PASS, FAIL or NOTRUN for each criterion id. The last recorded aggregate run, `tests/Alltest all` on 2026-10-06, gave 319 PASS, 0 FAIL and 1 NOTRUN (the v2606 compile criterion, run separately). Later stage-specific runs gave 78 PASS, 0 FAIL, 2 NOTRUN (M0, M9, M10c, including all nine GEMS gas-engine criteria) and 43 PASS, 0 FAIL, 13 NOTRUN (M0, M10a–d; the NOTRUNs are external-data studies not selected and the v2606 compile). Two later kernel repairs and the application cases were then covered by targeted checks and 19 Python tests of the benchmark scripts. The aggregate suite was re-run on the final sources on 2026-10-11/12 (plan section 39, item 7): 330 PASS, 0 FAIL and 3 NOTRUN, after six parallel pipe criteria that had stopped at the 30-minute MPI time limit on a heavily loaded machine passed with a longer limit. The NOTRUNs are the v2606 regression gate (passed in a separate v2606 run, 27 PASS), a bridge-provenance criterion that is NOTRUN by design, and a timing benchmark refused on the loaded machine. Criteria needing licensed Bi–I data are NOTRUN without the external files.

Selected criteria: the half-cell wall law converges at order 2.0 on a Graetz problem with a Robin wall (Péclet 5, Biot 1 and 50), with errors of the developed decay rate of $8 \times 10^{-7}$ to $2 \times 10^{-4}$; two independent pairs equal two single-pair runs byte for byte, serial and on four ranks; binary restarts are byte-identical; 1% deposition onsets of a 1000–350 K channel agree with an independent plug-flow equilibrium within 5.9 K (gate 10 K), and the five onsets of the reactive metal-vapour channel within 3.9 K for both BiI(g) data sets.

## 4.2 Conservation

| Quantity (relative to supplied material) | Production (19 runs) | Follow-up (60 runs) |
|---|---|---|
| Largest raw element residue $R_k$ | $1.41 \times 10^{-12}$ | $8.85 \times 10^{-13}$ |
| Largest accounted closure | $1.12 \times 10^{-15}$ | $9.60 \times 10^{-16}$ |
| Largest linear-solver defect | $1.45 \times 10^{-12}$ | $8.64 \times 10^{-13}$ |

The baseline logs report no gas-equilibrium failures and a maximum equilibrium element residue of $6.43 \times 10^{-16}$. These figures establish conservation of the discrete model, not physical accuracy.

## 4.3 Numerical studies

Table 2 lists the iodine-profile changes $E_{L1}$ (percent of the reference deposit, 1 cm bins) between successive meshes and schemes, and for halving the time step. Halving the 2 ms step changes profiles by 0.0043–0.3228%, and eight instead of three species correctors over a full 60 s run by at most 0.021%. The spatial error is larger and is dominated by first-order upwind diffusion: at equal mesh upwind differs from the bounded `limitedLinear` scheme by 7.7–20.8% in the most sensitive case, whereas two bounded schemes differ by at most 2.8%. On a fourth level (93,312 cells) that case changes by

**Table 2:** Iodine profile changes $E_{L1}$ (%) between successive mesh levels at 2 ms, relative to the finer level, and for halving the time step (18,432 cells, upwind). Steel cases are Q2 (Liu et al. steel column); the shifted-BiI case uses 1% metal saturation. A dash marks a combination not run. (Generated table of the methods manuscript.)

| Case | Scheme | 8,192→18,432 | 18,432→41,472 | 41,472→93,312 | 2→1 ms |
|---|---|---|---|---|---|
| Pure $PbI_2$ | upwind | 6.57 | 5.06 | – | 0.0043 |
| | limitedLinear | 5.70 | 2.15 | – | – |
| | van Leer | 4.94 | 2.23 | – | – |
| Steel, shifted BiI | upwind | 10.47 | 10.58 | 2.13 | 0.1536 |
| | limitedLinear | 15.35 | 8.95 | 1.98 | – |
| | van Leer | 16.83 | 9.17 | – | – |
| Steel Q2, 0.1% | upwind | 9.02 | 5.26 | – | 0.2326 |
| Steel Q2, 1% | upwind | 9.72 | 2.06 | – | 0.3228 |

1.98%, its total deposit by 0.015% and its centroid by less than 1 mm. Apparent orders (1.3 and 3.7) are inconsistent, so no formal grid convergence index is claimed. Production runs therefore use `Gauss limitedLinear 1` on 41,472 cells.

### 4.4 Equivalence of the gas engines (qualification channel)

Case `pbbi-gems-speciation` is a synthetic 1173–350 K channel with a prescribed $PbI_2/BiI_3$ source (iodine mole fraction $3.9 \times 10^{-6}$), 1 ms steps for 3 s, run with all three engines (`verification.json`, `qualification.json`):

| Quantity | Value |
|---|---|
| Deposit L1, frozen vs kernel, $PbI_2$ / $BiI_3$ (1 cm bins) | $1.03 \times 10^{-10}$ / $1.06 \times 10^{-7}$ (gate $10^{-3}$) |
| Deposit L1, local vs frozen | 0 (gate $10^{-9}$) |
| Maximum element closure | $6.66 \times 10^{-16}$ |
| Local GEMS3K calls; mean cost incl. both refinements | 5,814,420; 132.24 $\mu$s (gate 150 $\mu$s) |
| Reduced map (936 points): species error of refined result | $6.82 \times 10^{-10}$ |
| Reduced map: element error; raw IPM failures | $5.03 \times 10^{-16}$; 0 |
| Reduced map: points refused by the guards | 236 |
| Reduced map: largest raw IPM deviation (bulk $x = 10^{-10}$) | 0.64 decades |
| GEMS3K vs open-data $\log_{10} K_f$ | $\leq 3.69 \times 10^{-4}$ decades |
| Serial / four-rank closure; frozen restarts | $\leq 10^{-14}$; byte-identical |

The raw IPM deviations of up to 0.64 decades at trace dilution are the reason why mode local passes every IPM result through the exact kernel, which computes the final speciation (section 4.5). Injected fatal and NaN-dual failures are counted, re-create the engine where required, and leave the fields byte-identical to the frozen kernel fallback. These are numerical results under provisional inputs (synthetic carrier ramp and common diffusivity).

### 4.5 Production-scale per-cell GEMS3K run

The full production case of the Liu et al. steel experiment (Q2 source, 1% metal saturation; $576 \times 72 = 41{,}472$ cells, `Gauss limitedLinear 1`, 60 s at 2 ms) was repeated with the GEMS3K engines, changing only the `respeciation` engine (table 3). The bridge-enabled solver was built from the same solver sources as the production binary; run serially with the kernel engine it reproduces the production ledgers bit for bit. The GEMS3K runs used 8 MPI ranks (axial slabs, CPU-bound); an 8-rank kernel run with the same decomposition separates parallel round-off from the engine effect.

Using GEMS3K's standard-state energies in every re-speciated cell (19,944 distinct cell states; formation constants within $3.64 \times 10^{-4}$ decades of the kernel table) changes the iodine, $PbI_2$, $BiI_3$,

**Table 3:** Production-scale comparison of the gas engines (records in `run/024-liu2025/gems_engine_check/`).

| Quantity | kernel, serial | kernel, 8 ranks | GEMS frozen | GEMS local |
|---|---|---|---|---|
| Wall-clock time [h] | 25.04 | 6.37 | 6.74 | 10.73 |
| CPU time [core-h] | 25.0 | 50.9 | 53.9 | 85.8 |
| GEMS3K interior-point calls | – | – | – | $5.145 \times 10^8$ |
| Call failures / engine re-creations | – | – | – | 0 / 0 |
| Lower-guard activations (trace $< 10^{-12}$) | – | – | – | 14.0% |
| Mean cost per call [$\mu$s] | – | – | – | 150.8 |
| Largest element closure | $5.4 \times 10^{-16}$ | $5.5 \times 10^{-16}$ | $5.8 \times 10^{-16}$ | $5.8 \times 10^{-16}$ |
| Iodine profile $E_{L1}$ vs serial kernel | 0 | $1.8 \times 10^{-4}$ | $9.9 \times 10^{-5}$ | $9.9 \times 10^{-5}$ |
| $BiI_3$ share of iodide iodine [%] | 94.5837 | – | 94.5841 | 94.5841 |

**Table 4:** Q1 iodide maxima on the production mesh compared with the published peaks [2]. Subscript bin: 1 cm binned maximum; 3: maximum of the 3 cm smoothed profile; $T_c$: wall temperature at the deposit centroid. Positions in cm, temperatures in °C; experimental ranges are those printed in the paper.

| Case | Species | $x_{exp}$ | $x_{bin}$ | $x_3$ | $T_{exp}$ | $T_{bin}$ | $T_3$ | $T_c$ |
|---|---|---|---|---|---|---|---|---|
| Pure $PbI_2$, steel | $PbI_2$ | 36 | 36 | 37 | 368±18 | 366.5 | 350.5 | 348.7 |
| LBE, steel | $PbI_2$ | 51 | 51 | 52 | 288±18 | 288.1 | 269.2 | 274.5 |
| LBE, steel | $BiI_3$ | 57 | 57 | 58 | 154±14 | 166.9 | 138.8 | 158.1 |
| LBE, silica | $PbI_2$ | 51 | 56 | 56 | 306±24 | 287.8 | 287.8 | 288.8 |
| LBE, silica | $BiI_3$ | 59 | 63 | 64 | 127±39 | 152.8 | 119.8 | 147.1 |

Pb and Bi deposit profiles by at most $2.1 \times 10^{-4}$ (relative $L_1$ of the 1 cm profiles), the $BiI_3$ share of iodide iodine by 0.0004 percentage points and the peak temperatures by less than 0.003 K. Parallel round-off alone is of the same order (up to $2.7 \times 10^{-4}$); at equal decomposition the engine effect is at most $1.8 \times 10^{-4}$.

Modes frozen and local give bitwise identical results in all fields, profiles and ledger records. This follows from the implementation: in mode local the warm GEMS3K interior-point result starts a kernel solve, and the final speciation is then computed by the canonical cold kernel solve from the same GEMS3K thermodynamic data. The 514.5 million GEMS3K calls (none failed) therefore demonstrate the robustness and cost of per-cell GEMS3K calls at production scale (59% above the frozen engine), but they do not shape the result. The coupling as used at production scale is GEMS3K thermodynamics per cell with an exact, element-conserving equilibrium solve; a GEMS3K-solved equilibrium, and condensed phases inside GEMS3K, are not demonstrated.

# 5 Modelling applications

## 5.1 Liu et al. (2025): Pb–Bi–I deposition from LBE

Liu et al. [2] evaporated $PbI_2$ and iodine-containing LBE into helium in steel (4.8 mm) and silica (5.0 mm) columns. The model transports the nine Pb–Bi–I gases, with HKS exchange of $PbI_2$ and $BiI_3$ ($C_e = s_i = \alpha_i = 1$, half-cell resistance), the Pb and Bi metal channels and the reactive BiI channel, in a frozen carrier built from the digitised wall temperatures. 15 baseline and 64 follow-up cases (60 s; four flow-stop cases continued to 80 s) and 19 production re-runs were completed. Source Q1 releases $PbI_2$ and $BiI_3$ at the mean rate of the deposited masses, so the split is imposed; Q2 releases elements plus LBE metal vapour at an unmeasured fraction $\beta$ of saturation, so the split emerges.

Key results (table 4 and fig. 3):

- All five binned Q1 peak temperatures lie inside the printed ranges. The smoothed and centroid temperatures lie at or near the lower limits, and the silica maxima lie 5 and 4 cm downstream of the measured positions. The Q1 $BiI_3$ share of deposited iodide iodine (65.34% steel, 28.68%

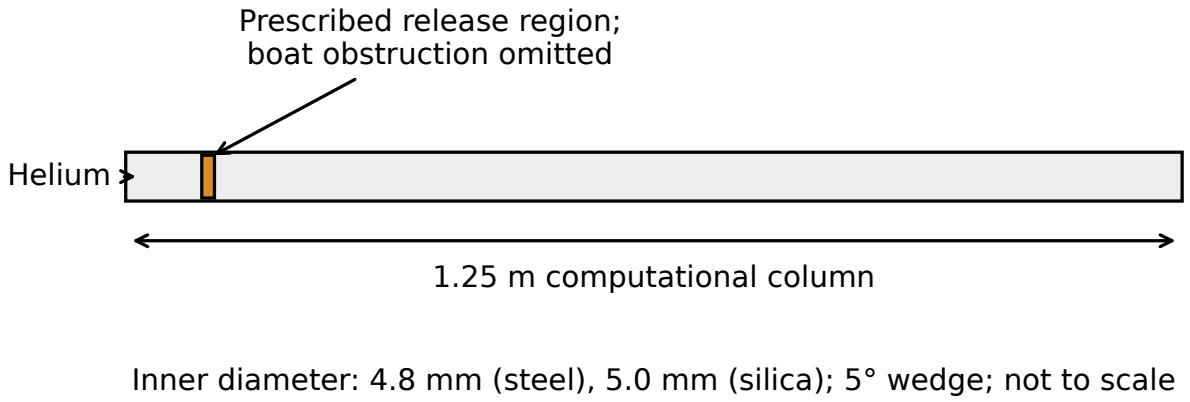



**Figure 2:** Computational geometry of the Liu et al. columns: a 1.25 m tube, represented by an axisymmetric 5° wedge, with a prescribed source region near the hot end. The boat itself is not resolved.

silica, against 65.63% and 29.03% measured) is conditioned on source masses taken from the same deposits and is not a prediction.

- The silica temperature curve and its printed peak labels disagree by about 63 K and 108 K. Two input-derived shifts of the curve (−4.10 and −4.65 cm) raise the profile overlap from 6.13% to 77.80% and 73.26%; they are hypotheses, not corrections, and their ranking reverses between advection schemes.
- **Main sensitivity: BiI(g) thermodynamics.** At 1% metal saturation in steel, the baseline (Barin) data give a $BiI_3$ share of 94.58%; raising the BiI(g) Gibbs energy by 29 kJ mol$^{-1}$ gives 2.01%. With this shift the open-data gas speciation comes close to Liu's published equilibrium figures (best shifts 29–30 kJ mol$^{-1}$; rms deviation 0.014–0.047 instead of 0.270–0.368). The shift is a sensitivity surrogate, not a qualified record. The split also depends on the unmeasured metal loading, so no branch is selected.
- Input sensitivities (iodine profile, 18,432-cell upwind differences): gas diffusivity up to 14.20%, flow reference conditions 5.09–10.93%, iodide accommodation up to 4.21%, source position at most 0.19%. Diffusion, not interfacial kinetics, limits wall exchange (wall Damköhler number $1.4 \times 10^3$–$2.9 \times 10^3$ at $\alpha = 1$).

### 5.2 Lobresco et al. (2026): $PbI_2$ head-to-head

The five $PbI_2$ experiments of Lobresco et al. [3] (helium flows 5, 106 and 540 mL/min, durations 6–300 s) were run without adjusting any parameter: HKS with $C_e = \alpha = 1$ and the half-cell resistance, the digitised published wall temperatures, an evaporating sample, 22,896 cells and limited-linear advection. With the paper's 1%-of-peak onset rule (table 5 and fig. 4) the onset temperatures differ from the measurements by +26, +19, +16, −11 and −14 K, an RMS of 18 K against 35 K for both published models; the two meshes agree within 0.2 K. The computed peaks lie 1–4 cm upstream of the measured ones. Not represented: the flow stop and withdrawal at the end of the experiment, the boat, and the non-dilute source region ($PbI_2$ mole fraction 0.33–0.36 next to the sample). This is an independent comparison with published measurements and models, not a validation.

## 6 Code availability and use

**Repository.** Solver, bridge, generators, benchmarks, cases and tests are in the LESTO repository, `https://github.com/JanKren/LESTO` **[TO CONFIRM: repository access, release tag or commit to be cited, and the licence of the repository's own code]**: `applications/rhoFixedFlowFoam` (solver, `tests/`, design and decision record `doc/phase-change-plan.md`),

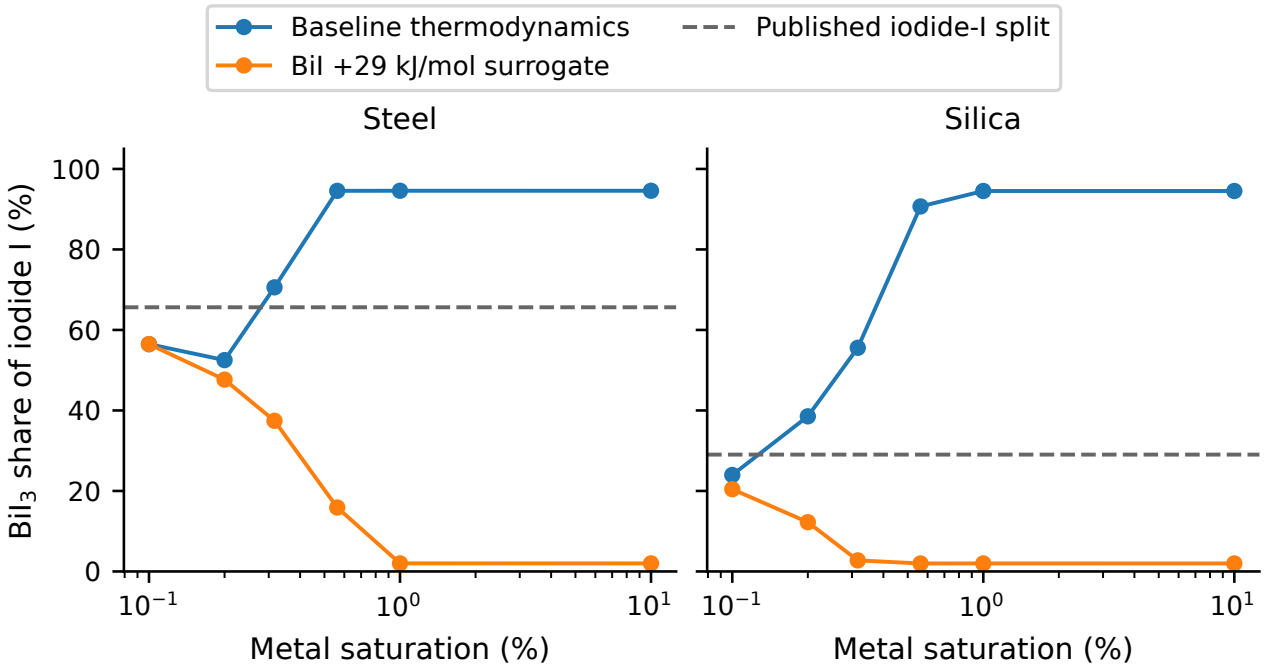



**Figure 3:** Conditional sensitivity of the $BiI_3$ share of deposited iodide iodine to metal loading $\beta$ and BiI(g) thermodynamics, for the steel and silica LBE cases (source Q2). Dashed grey lines: published $BiI_3$ share after excluding KI [2]. The +29 kJ/mol branch is a surrogate perturbation.

**Table 5:** Deposition-onset temperatures for the five $PbI_2$ experiments of Lobresco et al. [3]: flow (mL/min, standard), duration (s), measured onset (K), and onset minus measurement (K) for this work and for the temperature-based and vapour-pressure-based models of that paper. $\Delta x_{pk}$: computed minus measured peak (cm); $L_1/n_0$: absolute profile difference over the sample amount.

| Case | $\dot{V}$ | $t$ | $T_{exp}$ | This work | Temp.-based | Vap.-press. | $\Delta x_{pk}$ | $L_1/n_0$ |
|---|---|---|---|---|---|---|---|---|
| T106_6 | 106 | 6 | 814 | +26 | −39 | +27 | −4 | 0.89 |
| T106_60 | 106 | 60 | 723 | +19 | −10 | +37 | −2 | 0.66 |
| T106_300 | 106 | 300 | 667 | +16 | −1 | +34 | −2 | 0.88 |
| T5_60 | 5 | 60 | 847 | −11 | −60 | −50 | −1 | 0.37 |
| T540_60 | 540 | 60 | 700 | −14 | −30 | −15 | −1 | 0.47 |
| RMS | | | | 18 | 35 | 35 | | |

`thermochemistry/` (`gemsbridge`, `systems`, `benchmarks/liu2025`, `benchmarks/lobresco2026`), and `run/`.

**Case ladder.** The cases in `run/` add one capability at a time. Because they are being renumbered into a continuous ladder, they are cited here by slug:

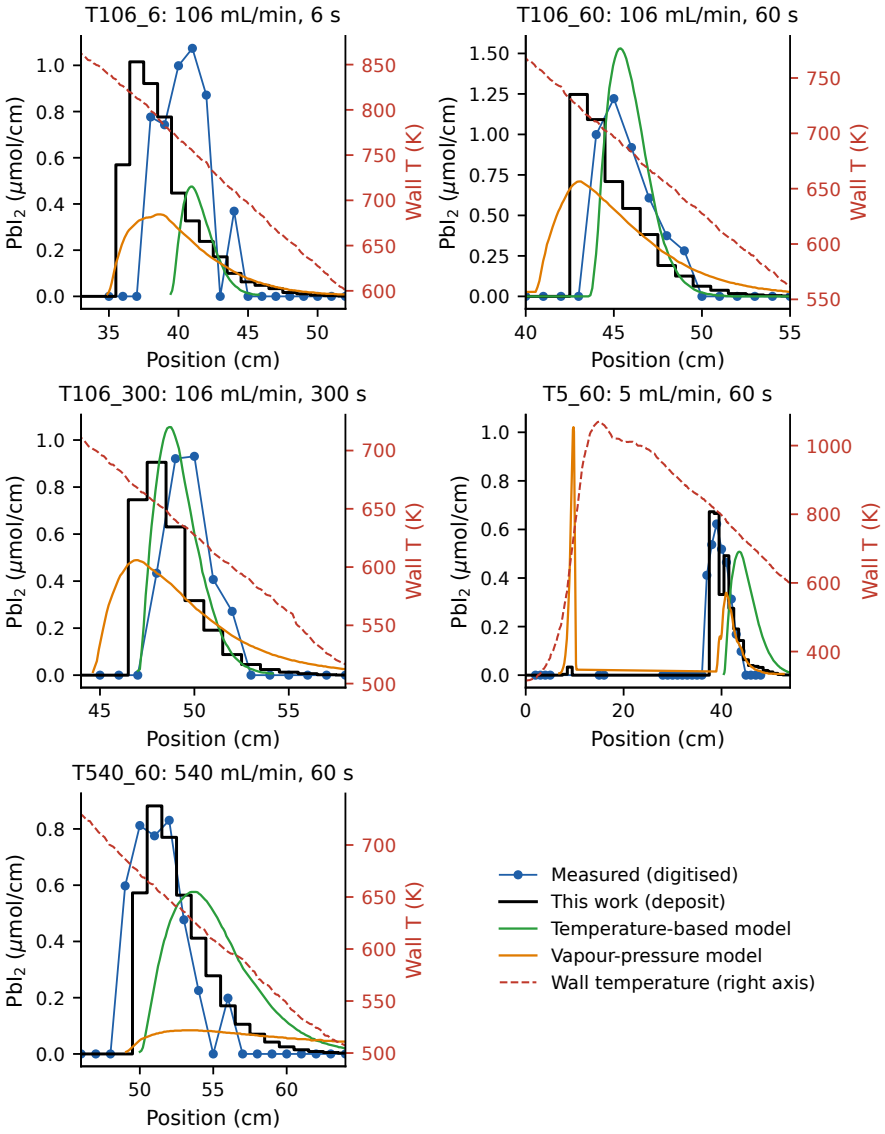



**Figure 4:** Deposited $PbI_2$ for the five experiments of Lobresco et al.: measured profiles and the two published model results (digitised, adapted under CC BY 4.0 [3]), and this work at the end of each experiment. Dashed red: digitised wall temperature.

| Slug | Capability |
|---|---|
| `first-pipe-flow-simpleFoam` … `variable-properties-high-temperature` | carrier development: laminar pipe, parabolic inlet, energy, variable wall temperature, density and properties |
| `one-species-transport-rhoFixedFlowFoam`, `variable-diffusivity`, `handles-solid-species`, `coupled-species-sources` | frozen-carrier transport, $PbI_2$–He diffusivity, condensed species, coupled sources |
| `temperature-model`, `paper-temperature-model` | temperature-based deposition law; paper-mode carrier |
| `hks-inventory`, `hks-gems-local` | HKS law with an evaporating inventory; $p_{eq}$ from GEMS3K (mode local, M9) |
| `pbi2-bii3-channel`, `pbbi-gas-speciation`, `pbbi-gems-speciation`, `lbe-metal-vapour` | two iodide pairs; kernel re-speciation; GEMS3K gas engines; metal vapour and reactive channels |
| `liu2025`, `lobresco2026` | the applications of section 5 |

**Building and running.** With OpenFOAM v2412 sourced:

- default solver: `applications/rhoFixedFlowFoam/Allwmake`;
- with GEMS3K: build the static GEMS3K and the bridge (`build-gems3k-static.sh`, `make && make check`, conda-forge toolchain), then `exportLESTO_GEMSBRIDGE=<LESTO>/thermochemistry/gemsbridge` and `Allwmake`; `rhoFixedFlowFoam -help` states whether the bridge is compiled in. The bridge is machine-specific (glibc $\geq 2.34$) and is rebuilt on each machine, e.g. on the PSI cluster Merlin7;
- cases: `./Allrun` in the case directory, with the external thermodynamic paths in environment variables (`LESTO_M10_SYSTEM`, `LESTO_M10_DATA`, `LESTO_M10E_NASA`, `LESTO_M10E_THERMOFUN`) and, for the Liu cases, explicit flow-reference conditions;
- tests: `tests/Alltest <milestones>` or `all`.

**Data licences.** GEMS3K is LGPL-3.0; the NASA CEA records and the files derived from them are Apache-2.0. The Bi–I data derive from HERACLES (ThermoHub, GPL-3.0; Barin [13]); their provenance and redistribution terms are unconfirmed, so no Bi–I record or derived table is in the repository and the Bi–I criteria and cases need the external files. JANAF-derived files are internal checks only and data derived from HSC Chemistry are not redistributable. The digitised curves of Liu et al. and Lobresco et al. are adapted under the source papers' CC BY 4.0 licences.

# 7 Limitations and outlook

**Limitations.**

- **Engine of record.** The reported results use the exact kernel; per-cell GEMS3K is qualified on a channel but not yet shown at production scale (section 4.5).
- **Physics not represented.** Frozen carrier (no buoyancy, latent heat or species feedback), ideal gas, no condensed-phase equilibrium in cells, no LBE source equilibrium, prescribed release, no aerosol formation, Soret diffusion, surface chemistry, KI, water or oxygen chemistry; the BiI wall channel is irreversible and phenomenological; instantaneous gas equilibrium replaces unmeasured kinetics.
- **Inputs.** The +29 kJ/mol BiI(g) value is a surrogate; the $BiI_3$ liquid record is constructed; the Bi data provenance is unqualified. Missing experimental inputs: original temperature profiles and scan coordinates, flow-controller reference conditions, release histories, boat geometry, surface coefficients.

- **Numerics.** The indicative spatial error of the profile on the production mesh is 2.5–3.6% for the most sensitive case; no formal mesh independence is claimed. The source region of the Lobresco cases is not dilute.

**Outlook.**

- Complete the production-scale GEMS3K run and the aggregate test re-run on the final sources.
- Obtain the missing experimental inputs and a qualified BiI(g) record before any predicted iodide split is used; clarify the Bi data licence.
- Extend GEMS3K coupling to phases where it adds information beyond the tables: non-ideal or multi-component condensates, and source equilibrium with liquid LBE once iodine activity data are available.
- Machine-learning and reduced-order surrogates of the coupled model are planned in a later phase of Task 3.4. The cost analysis above indicates that for the Pb–Bi–I gas alone a tabulated surrogate gains little over the cached exact solver, so surrogates are expected to matter for larger systems and for reactor-scale transport.

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